

Boyu Yao

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EDUCATION

Johns Hopkins University

Aug. 2023 – Present

Ph.D. in Civil and Systems Engineering

Advisor: Prof. Yury Dvorkin

Affiliations: Ralph O'Connor Sustainable Energy Institute (ROSEI); Critical Minerals Research Lab; Electric Power Innovation for a Carbon-free Society (EPICS); Center for Systems Science and Engineering (CSSE); Academic Center for Reliability and Resilience of Offshore Wind (ARROW)

Johns Hopkins University

Jan. 2021 – May 2022

Master of Science in Engineering, Geography and Environmental Engineering

Advisor: Prof. Ben Hobbs

Concentration: Environmental Management and Economics

University of California, Irvine

Aug. 2019 – June 2020

UCI 3+2 Program, Henry Samueli School of Engineering

Exchange Student, Civil and Environmental Engineering

Tianjin University

Sept. 2016 – June 2020

Bachelor of Engineering, Environmental Engineering

PROFESSIONAL EXPERIENCE

Scholar

Sep. 2026 – Jul. 2027

Critical Minerals Research Lab

Washington, DC & Remote

A research hub launched by Resources for the Future to approach critical minerals research with an interdisciplinary lens.

- Participate in the 2026–2027 research cohort, critique fellow research presentations, and engage in policy discussions and policy brief write-ups.

Research Intern

Jun. 2026 – Aug. 2026

Resources for the Future

Washington, DC

Supervisors: Karen Palmer and Jenya Kahn-Lang

- Conduct research on capacity markets, resource adequacy, and electricity market design by evaluating and comparing analytical models and decision-support tools.

Graduate Research Assistant

Aug. 2023 – Present

Johns Hopkins University

Baltimore, MD

Advisor: Prof. Yury Dvorkin, Smart Energy reseARCH (SEARCH) group

- **Clean-Energy Supply Chains:** Designed a national framework linking critical material supply chains to U.S. decarbonization targets and clean-energy deployment feasibility.
- **Supply-Chain-Constrained Capacity Expansion:** Built an optimization model to quantify how critical material, manufacturing, and land constraints shape generation capacity expansion.

- **Grid-Supporting Equipment Supply Chains:** Integrated stock-flow, bill-of-materials, supply-use, and expansion modeling to assess how equipment and material constraints shape power system expansion.
- **Maryland Energy Resilience and Efficiency:** As part of the Maryland Energy Resilience and Efficiency Working Group (EREWG), analyzed electricity storage pathways and grid transformation strategies to enhance resilience and support Maryland’s 2031 decarbonization targets.
- **Advanced Nuclear Integration to PJM:** In collaboration with Constellation, assessed Gen IV nuclear flexibility and PJM market impacts under rising data center demand using capacity expansion modeling.

Research Associate

Aug. 2022 – Aug. 2023

University of California, San Diego

San Diego, CA

Supervisor: Prof. Michael Davidson, Power Transformation Lab

- **Winter Resource Adequacy Analysis:** Conducted power system reliability analyses for the Northeast China Grid, providing insights into resource adequacy, seasonal demand, and renewable energy integration.
- **Interprovincial Electricity Trade Analysis:** Assessed the impact of long-term contracts on system costs and reliability in China’s power grid, proposing alternative designs through simulation-based analysis.

Research Assistant

Jan. 2022 – May 2022

Johns Hopkins University

Baltimore, MD

Advisor: Prof. Ben Hobbs

- Enhanced the JHSMINE energy optimization model by adding variables and constraints, updating system cost calculations, and automating data processing and visualization.

Consultant

May 2021 – Dec. 2021

Tsintergy Technology

Guangzhou, Guangdong, China

Energy Operations Research Center, Tsinghua University

- Analyzed Guangdong’s energy generation and market mechanisms and developed a carbon emissions calculation guide based on IPCC methodology.
- Supported renewable energy policy analysis and energy demand forecasting for Qinghai and Jiangsu through strategy assessment, MATLAB model enhancement, grey modeling, and emissions estimation.

RESEARCH PUBLICATIONS

- Saroj Khanal, Geon Roh, **Boyu Yao**, Abraham Silverman, Dennice Gayme, Charalambos Konstantinou, Jip Kim, Yury Dvorkin: *Shift or Curtail? How Much Data-Center Flexibility Is Worth Depends on the Host Power Grid*. Manuscript under review; *arXiv preprint*, August 2026.
- **Boyu Yao**, Yury Dvorkin: *Grid-Supporting Equipment Supply Chains Constrain the Feasible Pace of Power System Expansion*. Manuscript under review; *arXiv preprint*, April 2026.
- Owen Reed, **Boyu Yao**, Saroj Khanal, Magdalena Klemun, Yury Dvorkin: *Electricity Prices under Rapid Load Growth: The Case of Data Centers in Virginia*. *Oxford Energy Forum*, Mar. 2026.
- Ming Wei, **Boyu Yao**, Michael Davidson: *Resource Adequacy Under Institutional Constraints and the Low-Carbon Energy Transition in China*. *Energy Policy*, Mar. 2026.
- **Boyu Yao**, Andrey Bernstein, Yury Dvorkin: *Generation Expansion Planning with Upstream Supply Chain Constraints on Materials, Manufacturing, and Deployment*. *IEEE Transactions on Energy Markets, Policy, and Regulation*, Jan. 2026.
- **Boyu Yao**, Hannah Jeong, Mahdi Mehrtash, Bentley Allan, Daniel Ockerman, Yury Dvorkin: *Understanding Supply Chain Constraints for the U.S. Clean Energy Transition*. *Nature Portfolio Journal Clean Energy*, Sep. 2025.

- **Boyu Yao** and Yury Dvorkin: *A Decision Framework for Clean Energy Supply Chain Optimization*. In *Proceedings of the 2025 IEEE Power & Energy Society General Meeting (PESGM)*, Austin, Texas, USA, July 2025.

TECHNICAL REPORTS, POLICY BRIEFS & DATASETS

- **Boyu Yao**, Abe Silverman, Yury Dvorkin : *The Next Data Center Bottleneck Is Grid-Supporting Equipment*. Policy brief for the Ralph O'Connor Sustainable Energy Institute (ROSEI), Johns Hopkins University. Publicly accessible at ROSEI Policy Brief, April 2026.
- Owen Reed, Saroj Khanal, **Boyu Yao**, Dyer Combariza, Magdalena Klemun, and Yury Dvorkin: *ICARUS Data Center Dataset (ICARUS-DC)*. Versioned GitHub repository, May 2026. Versioned GitHub repository.
- Saroj Khanal, **Boyu Yao**, Marisel Villafañe-Delgado, and Yury Dvorkin: *ICARUS PJM Dataset (ICARUS-PJM)*. Versioned GitHub repository, Mar. 2026. Versioned GitHub repository.
- Yury Dvorkin, Benjamin Hobbs, Ziting Huang, Stephanie Wilcox, **Boyu Yao**: *Energy Resilience and Efficiency in Maryland*. Technical report for the Energy Resilience and Efficiency Working Group, State of Maryland. Publicly accessible at Maryland EREWG, Dec. 2024.
- Ming Wei, **Boyu Yao**, Chi Gao, Max Dupuy, Michael Davidson: *North China and Northeast China Inter-provincial Electricity Trade Brief*. Publicly accessible at Electricity Market Tracker, Oct. 2023.
- Fikri Kucuksayacigil, **Boyu Yao**, Ming Wei, Chi Gao, Max Dupuy, Michael Davidson: *Northeast China Grid Winter Resource Adequacy Report*. Publicly accessible at Electricity Market Tracker, July 2023.

SELECTED PRESENTATIONS

- **(Invited)** *Beyond Market Signals: Supply-Chain Constraints in Grid Expansion*. 2026 Power Markets Journal Club.
- **(Poster)** *The ICARUS Dataset: An Infrastructure-Centric, Open Dataset for Capacity Expansion, Resource Adequacy, and Data Center Integration*. 2026 Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit.
- **(Poster)** *When the Supply Chain Shapes Long-Term Grid Planning*. 2026 Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit.
- **(Poster)** *When the Supply Chain Shapes Long-Term Grid Planning*. 2025 The Electric Power Innovation for a Carbon-free Society (EPICS) Center US Summit.
- **(Invited)** *Market Signals and Incentives Aren't Sufficient: The Case for Supply-Chain Modeling for Grid Expansion*. 2025 INFORMS Annual Meeting.
- **(Oral)** *When the Supply Chain Shapes the Grid: Integrating Upstream Constraints into Long-Term Planning*. JHU Energy and Environment Systems Seminar, 2025.
- **(Oral)** *Integrating Upstream Supply Chains into Generation Expansion Planning*. JHU Center for Systems Science and Engineering (CSSE) Meeting, 2025.
- **(Poster)** *A Decision Framework for Clean-Energy Supply-Chain Optimization*. 2025 IEEE Power & Energy Society General Meeting.
- **(Poster)** *Understanding Supply Chain Constraints for the U.S. Energy Transition*. 2025 JHU Sustainability Symposium on Research, Practice, and Education.
- **(Poster)** *Understanding Clean-Energy Supply Chains: A U.S. Perspective*. 2025 Ralph O'Connor Sustainable Energy Institute (ROSEI) Summit.
- **(Oral)** *Energy Resilience and Efficiency Analyses to Support the Maryland Climate Change Commission*. Annual Meeting of the Energy Resilience and Efficiency Working Group, Maryland Department of the Environment, 2025.

- **(Invited)** *A Fully Open-Source Power Dataset for Policy Applications in China*. 2023 INFORMS Annual Meeting.
- **(Oral)** *Power System Resource Adequacy in the Low-Carbon Transition—Northeast China Grid Winter Resource Adequacy*. 30th Technical Workshop of the China Energy Modeling Forum (CEMF).

TEACH & MENTOR EXPERIENCE

Teaching Assistantship

- **Energy Systems** (Prof. Yury Dvorkin, JHU, Aug. 2024 – Dec. 2024; Aug. 2025 – Dec. 2025; Aug. 2026 – Dec. 2026): Led weekly office hours and graded assignments/exams for over 30 students; supported student learning in Unit Commitment, Economic Dispatch, and Optimal Power Flow.
- **Energy Policy and Planning Models** (Prof. Ben Hobbs, JHU, Jan. 2022 – May 2022): Graded assignments/exams for over 20 students and facilitated review sessions on energy policy and planning models.

Research Mentorship

- **Luca Montoya** (JHU, ROSETAS, May 2025 – Aug. 2025): Research Topic: Resource Adequacy Assessment for Enriching Capacity Expansion Decisions.
- **Princeton Nguyen** (JHU, ROSETAS, May 2025 – Aug. 2025): Research Topic: Weather-Informed Scenario Generation for Capacity Expansion Planning in ISO-NE Under Renewable and Demand Uncertainty.
- **Maria Vabson** (JHU, Sep. 2023 – May. 2024): Research Topic: Critical Material Analysis in Clean-Energy Supply Chains: Identification, Stock Assessment, and Production Metrics.

SERVICE

Professional Service

- **Peer Reviewer** (Ongoing): Conducted peer reviews for academic journals, including *IEEE Transactions on Energy Markets, Policy and Regulation*; *IEEE Access*; and *Mitigation and Adaptation Strategies for Global Change*.

University & Departmental Service

- **Graduate Orientation Mentor** (JHU, CaSE, 2024 – Present): Guided incoming graduate students through departmental orientation and provided ongoing peer support to assist with their transition into the Civil and Systems Engineering program.

Volunteer Service

- **Primary School Science Teacher** (Chiang Mai, Thailand, Oct. 2017): Taught general science concepts to primary school students, developing lesson plans and facilitating engaging classroom activities.

AWARDS & HONORS

- **IEEE PES General Meeting Travel Fellowship** (*IEEE Power & Energy Society*) **2025**
- **Joseph Meyerhoff Fellowship** (*Johns Hopkins University*) **2023**
- **The 20th “Students of Science & Technology Outstanding Talents”** (*Tianjin University*) **2019**
- **Youth “Eco-Innovation Green Action” Competition** (*National, Third Prize*) **2019**
- **The 15th “Challenge Cup” National Undergraduate Curricular Academic Science & Technology Competition** (*Provincial, First Prize*) **2019**

- **The 12th “Challenge Cup” National Undergraduate Curricular Academic Science & Technology Competition** (*Tianjin University, First Prize*) **2019**
- **The 4th Tianjin College Students Environmental Discipline Innovation & Practice Invitational Tournament** (*Provincial, Second Prize*) **2018**
- **Representative of 3rd Plenary of the Fifth Council, Chinese Society for Engineering Education (CSEE)** (*“Integration of New Engineering Construction with Industry and Education” Int’l Conference*) **2018**
- **Merit Student** (*Tianjin University*) **2017**